Monitoring, Alerts & Water Conservation

Project: Household Water-Usage & Billing Monitor

Subject: Java Programming

Module 5

Project-Based Learning (PBL)



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Introduction

This module monitors household water consumption and helps users understand their usage patterns.



Tracks water consumption.



Compares usage across billing periods.



Identifies unusually high consumption.



Encourages responsible water usage.

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MONITORING

Water Usage Monitoring

Monitoring helps users understand how much water they consume.

Records household water usage.

Displays consumption for each billing period.

Compares current usage with previous usage.

Highlights changes in consumption.



20	35

Example: If usage increases from 20 to 35 units, the application can highlight the increase.

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THRESHOLD MONITORING

Alerts & Notifications

Alerts inform users when water consumption exceeds a configured limit.



Sets a usage threshold.



Compares recorded usage against that threshold.



Displays a warning for unusually high usage.



Encourages users to check for possible leaks or changes in habits.

30

35

Example: If the configured limit is 30 units and consumption reaches 35 units, the application displays a high-usage alert.

Note: This is an alert based on recorded data, not automatic proof of a leak.

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END-TO-END PROCESS

Complete System Workflow

All five modules work together to form one system.

01

Register customer details.

02

Record and validate meter readings.

03

Calculate water consumption and the bill.

04

Store billing records and generate reports.

05

Monitor usage and display alerts.

Workflow: Customer → Meter Reading → Bill Calculation → Database → Monitoring

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QUALITY CHECKS

Testing & Validation

Testing checks whether the application works correctly under different conditions.

- Test valid customer registration.
- Test incorrect meter readings.
- Verify bill calculations.
- Check database storage and retrieval.
- Verify high-usage alerts.
- Test error handling when the database is unavailable.

Purpose: Improve accuracy, reliability and usability.

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VALUE AND NEXT ITERATION

Benefits & Future Enhancements

Benefits

- Reduces manual billing work
- Organises customer and billing records.
- Helps users understand consumption.
- Encourages water conservation.

Future Enhancements

- Graphical dashboards and charts.
- Automated email or SMS alerts.
- Secure login for customers.
- Integration with smart water meters.

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MODULE 5

Conclusion

- The Household Water-Usage & Billing Monitor provides an organised way to record consumption, calculate bills and review usage.
- Uses Java programming and OOP concepts.
- Supports water-bill calculation.
- Maintains records using database connectivity.
- Helps users monitor consumption.
- Promotes responsible water usage.

Thank You!

Questions?



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